AVINASH KARKADA ASHOK

Biotechnology | Bioinformatics/Computational Biology

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Computational biologist with 4+ years of experience spanning NGS pipeline engineering, immune profiling, structural biology, and applied machine learning. Proficient in Python, R, Bash, Snakemake, and Docker on Linux/HPC and cloud platforms. Experienced in both academic research (Johns Hopkins, Indian Institute of Science) and industry (Infinity Bio, Sciomics), with a track record of delivering production-grade pipelines, reproducible analyses, and client-facing bioinformatics tools. 12+ peer-reviewed publications.

EDUCATION

M.S. Bioinformatics

Johns Hopkins University

Aug 2024 – May 2026 (exp.) | Baltimore, USA

B.E. Biotechnology

Visvesvaraya Technological University

June 2019 – July 2023 | Karnataka, India

First Class with Distinction

EXPERIENCE

Bioinformatics Engineer

Johns Hopkins University School of Medicine

Sep 2024 – Present | Baltimore, USA

- Engineer and maintain computational pipelines for InFlux, a highly multiplexed NGS-based influenza A neutralization assay that generates 50 strain-specific neutralization measures per sample, including pipeline validation and QC workflows to ensure data quality across diverse patient cohorts in **Dr. Benjamin H. Larman's** lab.
- Independently lead a **PhIP-Seq** antibody-profiling project analyzing several hundred serum samples investigating microbial dysbiosis and humoral immunity in early-onset, late-onset colorectal cancer, and polyp/healthy control cohorts; anticipated first-author manuscript in progress.

Summer Bioinformatics Intern

Infinity Bio, Inc.

June 2025 – August 2025 | Baltimore, USA

- Benchmarked and optimized NGS-based **MIPSA** analysis pipelines by evaluating alignment strategies (Bowtie2, Minimap2, etc.), improving alignment quality and I/O handling while reducing overall runtime and resource usage.
- Co-developed EpitopeFindeR 2, a reproducible pipeline integrating FoldDisco-based 3D structural mapping to identify continuous and discontinuous epitopes from library peptide sequences; pipeline deployed to production for client-facing data portal.
- Built **Latch Bio**-based analysis pipelines including a UMAP-based sequence-space analyzer and visualization module, improving customer experience for EpitopeFindeR 2 users.

Junior Research Fellow

Indian Institute of Science

Nov 2023 – Aug 2024 | Bangalore, India

- Co-first-authored a comparative genomics study decoding mammalian adaptive signatures in clade 2.3.4.4b H5N1 across non-human hosts, identifying convergent mutations that may increase human pandemic risk (published in *Microbiology Spectrum*).
- Demonstrated that neuraminidase stalk length and haemagglutinin glycosylation patterns molecularly bias segment pairing, elucidating reassortment routes that facilitated emergence of highly pathogenic clade 2.3.4.4b A(H5N1).
- Contributed to the discovery and characterization of RHIM motifs in bat RNA viruses including a SARS-CoV-2 Nsp13 RHIM, revealing species-specific ZBP1–RIPK3 cell death signaling (published in *Cell iScience*).

SKILLS

Programming & Platforms

Python | R | Bash | SQL | Java | Git/GitHub | Linux | HPC/SLURM | AWS (EC2, S3, GPU) | Docker | Singularity | Conda/Mamba | Jupyter | RStudio | Quarto | Shiny

NGS & Omics Analysis

PhIP-Seq | RNA-seq | scRNA-seq | ChIP-seq | WGS | WES | MIPSA | Bowtie2 | BWA | Minimap2 | STAR | HISAT2 | Kallisto | Salmon | SAMtools | BCFtools | BEDTools | GATK | Variant Calling | FastQC | Trimmomatic | Cutadapt | SRA Toolkit | IGV | DESeq2 | edgeR | limma | Seurat | Scanpy | Bioconductor | Biopython | BLAST | CD-HIT | HMMER | MAFFT

Workflow & Reproducibility

Snakemake | Latch Bio | Docker | Singularity | Git/GitHub

ML, Statistics & Visualization

scikit-learn | XGBoost | SVM | LASSO | Random Forests | PCA | UMAP | t-SNE | pandas | NumPy | SciPy | matplotlib | plotly | ggplot2 | GraphPad Prism

Structural Biology & Drug Discovery

GROMACS | AMBER | gmx_MMPBSA | AlphaFold | ESMFold | Rosetta | FoldX | HADDOCK | AutoDock | Maestro/Schrödinger | RDKit | PyMOL | UCSF ChimeraX

Wet Lab Techniques

Bacterial cell culture | PCR | Gel electrophoresis | SDS-PAGE | Western blot | ELISA | Sample handling & QC

RELEVANT PUBLICATIONS

- **Decoding non-human mammalian adaptive signatures of clade 2.3.4.4b H5N1 to assess its human adaptive potential**
  R. Nataraj, **A. K. Ashok**, A. A. Dey, and S. Kesavardhana
  Microbiology Spectrum.
- **High-throughput screening and molecular dynamics simulations of natural products targeting LuxS/AI-2 system as a novel antibacterial strategy for antibiotic resistance in *Helicobacter pylori***
  A. K. Ashok, T. S. Gnanasekaran, H. S. S. Kumar, K. Srikanth, N. Prakash, and P. Gollapalli
  Journal of Biomolecular Structure and Dynamics.
- **Bat RNA viruses employ viral RHIMs orchestrating species-specific cell death programs linked to Z-RNA sensing and ZBP1–RIPK3 signaling**
  S. Mishra, D. Jain, A. A. Dey, S. Nagaraja, M. Srivastava, O. Khatun, K. Balamurugan, M. Anand, **A. K. Ashok**, S. Tripathi, *et al.*
  Cell iScience.
- **Computational-driven discovery of AI-2 quorum sensing inhibitor targeting the $5'$-methylthioadenosine/S-adenosylhomocysteine nucleosidase (MTAN) to combat drug-resistant *Helicobacter pylori***
  M. Kumar, **A. K. Ashok**, T. Bhat, K. Ballamoole, and P. Gollapalli
  Computers in Biology and Medicine.

ACHIEVEMENTS & SERVICE

- Peer reviewer for *Nature Communications*, *Journal of Crohn's and Colitis*, *J. Biomolecular Structure and Dynamics*, and *J. Biological Macromolecules*.
- 12+ peer-reviewed publications as first- or co-author across influenza genomics, viral cell-death signaling, and computational drug discovery.
- Paper presentation at Immunology 2025 (AAI Annual Meeting).
- Awarded Best Paper Presentation (JAIVIK 2022) and Best Undergraduate Thesis.
- Mentored multiple students in practical bioinformatics, molecular dynamics, and NGS analysis.